

Question	Answer	Marks
2(a)	in (rotational) equilibrium	B1
	<u>sum / total</u> of clockwise moments about a point = <u>sum / total</u> of anticlockwise moments about the (same) point.	B1
2(b)	$80 \times 9.81 \times 3$ or $60 \times 9.81 \times 3$ or $45 \times 9.81 \times x$	C1
	$80 \times 9.81 \times 3 = (60 \times 9.81 \times 3) + (45 \times 9.81 \times x)$	C1
	$x = 1.3 \text{ m}$	A1
2(c)(i)	$k = F / x$	C1
	$x = 0.80 - 0.59$ $= 0.21 \text{ m}$	C1
	$k = (60 \times 9.81) / 0.21$ $= 2800 \text{ N m}^{-1}$	A1

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2(c)(ii)	$E = \frac{1}{2} kx^2$ or $E = \frac{1}{2} Fx$ or $E = \frac{1}{2} F^2/k$ $= \frac{1}{2} \times 2800 \times 0.21^2$ or $= \frac{1}{2} \times 60 \times 9.81 \times 0.21$ or $= \frac{1}{2} \times (60 \times 9.81)^2 / 2800$	C1
	$= 62 \text{ J}$	A1

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